

Research Papers

Probabilistic lithium-ion battery state-of-health prediction using convolutional neural networks and Gaussian process regression

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ABSTRACT

As the world accelerates its renewable energy transition, lithium-ion batteries are increasingly being used for large scale energy storage. Due to the high cost of production and replacement of large lithium-ion battery packs, accurate estimation of battery state-of-health (SOH) has become important. Accurate estimation is especially important for microgrid and transportation industries where operational batteries will be charged intermittently and incompletely. Incomplete charging scenarios provide models with inconsistent inputs, hence, cause inconsistent output accuracies, which are important to quantify in the context of dispatch controllers and predictive maintenance. In this paper, we present a model that provides a probabilistic prediction of battery SOH from periodic charging events as would be encountered in real world operations. The proposed model combines the feature recognition and many-to-one mapping abilities of the classic convolutional neural network (CNN) with the probabilistic characteristics of Gaussian Process Regression (GPR) models. The combined CNN-GPR model predicts battery SOH to less than 1% mean absolute error using random partial charge segments as input and displays its confidence in the prediction using the variance from the GPR. Additionally, the CNN-GPR is able to predict battery SOH irrespective of battery usage history; this is demonstrated by validation on dynamic data sets where the model achieves a mean absolute error of less than 1% despite only being trained on standardized degradation cycles.

1. Introduction

Lithium-ion batteries degrade based on complex interactions between external and internal conditions, namely temperature, current, voltage, and battery usage, where the battery usage can differ widely depending on the application mean state-of-charge (SOC), depth-of-discharge (DOD), and C-rates. Since most realistic usage conditions can only be approximated, it is of great interest to accurately estimate battery health independently of its usage.

Unfortunately, many of the methods used to estimate battery health are dependent on battery usage. In [1] Tian argues that all methods, both dependent on and independent of battery usage, can be classified into four main categories: model-based methods, hybrid methods, other methods, and data-driven methods. Some examples of model based methods are presented in [2,3], and [4] which leverage models built from a physical foundation such as electrochemical, equivalent circuit, or semi-empirical derivation methods. The model-based methods provide sufficient accuracy; unfortunately, they rarely provide methods for adapting to the change in their parameters, as batteries degrade and do not adapt well to varying battery usage. To improve the accuracy

of model-based methods, some have combined them with data-driven methods in an attempt to harness the benefits of both methods. In [5] an empirical model is derived from first principles and combined with a back-propagation neural network (BPNN) to predict and compensate for the error of the empirical model. The model achieves a mean absolute error (MAE) of approximately 2%, however, it still relies on full charge cycles for data processing and is not validated on a dynamic data set. Nevertheless, hybrid methods show promising abilities to achieve more accurate results than model-based methods, while being more robust to unknown inputs than purely data-driven methods. Unfortunately, the model-based portion of the model may still require specialized tests to fit model parameters which can be time consuming and difficult to perform accurately. Another offshoot of hybrid methods classified as “other” by Tian, contains the popular incremental capacity analysis (ICA) method, which attempts to extract degradation features from a battery’s capacity-voltage curve by calculating the derivative of capacity with respect to voltage. In [6], battery capacity is predicted to within 1% using ICA and support vector regression (SVR) on partial charging data. The SVR in [6] is used to reduce the method’s sensitivity

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to the range of partial charging data used, however, it is still more susceptible to noise than other methods, undermining its performance in practical applications.

The last category explored is data-driven methods. Data-driven methods have become increasingly popular because they provide the opportunity to accurately model the relationship between observed measurements and battery SOH without the need to fully model or understand the internal workings of the battery. This is especially useful for the modeling of lithium-ion batteries because of their complex side reactions and non-linear degradation behavior [7]. Consequently, there are many studies demonstrating the success of different machine learning algorithms such as support vector machines [8], temporal convolutional networks [9], feed-forward neural networks and long-short-term-memory (LSTM) [10], Gaussian process regression (GPR) [11,12], and convolutional neural networks (CNN) [10,13,14]. Although all of the mentioned methods are intriguing in their own right, the ability of the CNN to reduce multiple contemporaneous features down to a single output makes it well suited for the case where multiple time-series inputs (i.e. V , I) are being mapped to a single SOH value. This CNN characteristic is apparent in [13] where a CNN is used to predict battery SOH using partial voltage, current, and temperature charge profiles as input. The model achieved a MAE of less than 2% with noise added to the data to demonstrate model robustness. However, the practicality of this model is undermined by the need to train a new model for every SOC range that is used as input. In [14] a CNN is used with partial voltage and current charge data to predict battery capacity. The partial charges in this case all still contain the constant voltage portion of the charge data, requiring users to fully charge their device to 100% SOC to use the model. Additionally, the method in [14] requires new models to be trained for every charge segment that is used as data. In [15,16] a CNN is combined with the LSTM architecture to better predict battery remaining-useful-life. In [17] a CNN is combined with a gated recurrent unit network to improve the estimation of battery SOH. The combination of the CNN with network architectures that have memory ability improves the prediction accuracy and reduces training time [16]; however, the methods in [15,16], and [17] fail to represent the difference in estimation accuracy when using partial, instead of full, charges as model input.

In general, papers using the presented machine learning methods are absent of a metric that illustrates how confident the model is in its prediction. Probabilistic prediction is demonstrated in [12] where GPR is used to predict battery capacity purely from partial voltage charge data. The voltage data is segmented based on a lower and upper voltage bound so that they are independent of SOC. Unfortunately, this means that the model is only trained for a specific voltage segment. In [11] a sparse GPR method is used to predict battery SOH from capacity change. The capacity change is calculated from partial segments of the voltage vs capacity curve. The drawback to this approach is that the partial segments do not account for unique charging times and the sparse GPR requires a significant number of inducing points (500) to produce a sufficiently accurate prediction.

The stated drawbacks of the discussed papers are addressed in this paper by aiming to provide a model that is easily implemented, practical, and provides probabilistic predictions. To ease implementation, a CNN is leveraged to extract features from unprocessed charging data. To improve practicality, the CNN is trained on partial charge segments that have a random beginning SOC between 15 and 85 percent and a random charge length between 30 and 115 min at 0.3C; partitioning the data in such a way ensures that the model does not require a consistent portion of charge data to predict the battery SOH, as would be the case for real operational usage cases. Finally, sparse GPR is used to operate on the CNN predictions to capture the variation in CNN prediction accuracy from the inconsistent input characterization charge cycles and produce a meaningful, error bound, result for the user. By operating on the CNN output, rather than directly on the input, the number of required inducing points in the sparse GPR is greatly reduced from

hundreds down to just 20, improving computational efficiency, and therefore practicality of the model while maintaining the benefits of the GPR through its probabilistic predictions. Such a model will be useful in electric fleet management where vehicle charging is inconsistent, resulting in partial charges at different ranges of SOC. Because different ranges of SOC will provide different prediction accuracies, providing error bounds on the prediction will enable fleet managers to make more confident decisions by reliably identifying ESSs that are out of normal ranges within the fleet. Additionally, providing error metrics on the SOH prediction will allow battery management systems to operate more accurately under partial charging conditions as present in microgrid applications requiring real time dispatch controllers.

The remainder of this paper describes the test setup and data collection in Section 2, the Data-Driven CNN in Section 3, the CNN-only model tuning in Section 4, the models relationship to the data through integrated gradients in Section 5, the method for sampling random partial charge cycles in Section 6, and finally a discussion of the combined CNN-GPR in Section 7 followed by conclusions.

2. Data collection and experiment design

The data used for training and validation in this paper is from experiments performed at the Pacific Regional Institute of Marine Energy Discovery (PRIMED) Battery Lab. The batteries used are the Mollicel P42A [18] with parameters shown in Table 1.

The battery cycling was performed by an Arbin LBT 60 A, 16 channel, cell testing cabinet and the ambient temperature control was performed by a Thermotron S-8-8200 environmental chamber. The cycling of the 16 batteries, 1 per channel, followed the same general schedule of 14 days of degradation cycling intersected with a characterization cycle. Twelve of the sixteen channels were cycled with constant current charge/discharge cycles with different DOD_i and C_i . Two of the remaining channels were cycled with a dynamic degradation cycle that was designed to approximate the charge discharge pattern of an ESS connected to a solar array. The final two channels were calendar ageing cycles, hence, maintained a rest state for the degradation cycle. The characterization, static degradation, and dynamic degradation cycles are described in the following subsections.

2.1. Characterization cycle

The characterization cycle of the battery tests were run every 14 days. The tests followed the schedule outlined below where CC refers to constant current and CV refers to constant voltage and subscript i refers to channel number.

1. Set chamber temperature to $T_a = 23^\circ\text{C}$
2. 4 h rest (temperature soak)
3. CC discharge at C-rate = $-1/3$ until minimum voltage
4. 1 h rest
5. CC charge at C-rate = $1/3$ until maximum voltage
6. CV charge at maximum voltage to cutoff current

$$I = \frac{\text{NominalCapacity}}{100}$$
7. 1 h rest
8. CC discharge at C-rate = $-1/3$ until minimum voltage
9. 1 h rest
10. CC charge at C-rate = $1/3$ until

$$\int_0^t i(t) dt = (\text{Step 8 DischargeCapacity})/2$$
11. 1 h rest
12. EIS test from 100 kHz–1 Hz
13. 10 min rest
14. Direct current internal resistance test
15. 10 min rest
16. CC charge at C-rate = $1/3$ until

$$\int_0^t i(t) dt = (\text{Step 8 DischargeCapacity}) * DOD_i/2$$

Table 1
Battery manufacturer specifications.

Type	Chemistry	Form	Nom. Capacity (Ah)	Max Voltage	Min Voltage
Mollicel P42A	NCA	21 700	4.0	4.2	2.5

Table 2
DOD and C-rate values for each channel in the battery tests. A channel connects to a single cell.

Channel	C-rate	DOD
1	-4C	0.4
2	-1C	0.8
3	-2C	0.8
4	-1C	0.4
5	0C	0
6	Dynamic	Dynamic
7	Dynamic	Dynamic
8	0C	0
9	-2C	0.8
10	-4C	0.4
11	-4C	0.8
12	-2C	0.4
13	-1C	0.8
14	-1C	0.4
15	-2C	0.4
16	-4C	0.8

2.2. Standard degradation cycle

The degradation cycle ran uninterrupted for 14 day periods. The tests followed the schedule outlined below.

1. Set chamber temperature to $T_a = 35^\circ\text{C}$
2. 4 h rest (temperature soak)
3. CC charge at 1C to maximum voltage
4. CV charge at maximum voltage to cutoff current $I = \frac{\text{NominalCapacity}}{20}$
5. 20 min rest
6. CC discharge at C_i until $\int_{t_0}^t i(t) dt = (\text{NominalCapacity}) * DOD_i$
7. 20 min rest
8. Return to step 3

Where C_i and DOD_i correspond to the C-rate and DOD in Table 2 for each channel and $i(t)$, t_0 , and t refer to the measured current, start time, and end time of a charge step. Where 1C in Table 2 is equal to 4.2 A.

2.3. Dynamic degradation cycle

The test schedules for the dynamic degradation cycle were generated from a current time series designed to be representative of an ESS for a solar resource, while still being aggressive enough to increase battery degradation and reduce test times. The current direction (charge or discharge), is based upon the assumed probability that a system will be charging given what time of day it is. This is governed by the equation,

$$P_{\text{charge}} = \begin{cases} \frac{h}{12}, & \text{if } h \leq 12 \\ \frac{-h}{12} + 2, & \text{otherwise} \end{cases}$$

where $0 < h < 24$. To introduce randomness into the charging and discharging, a number, x , is sampled from a uniform probability distribution between 0 and 1 and then compared to a linear transformation of the probability P_{charge} . P_{charge} is transformed such that its limits are between 95 and 5%. Consequently, the final current direction is given by:

$$\text{current direction} = \begin{cases} 1, & \text{if } x \leq 0.9 * P_{\text{charge}} + 0.05 \\ -1, & \text{otherwise} \end{cases}$$

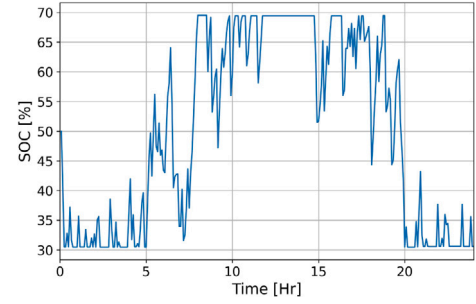


Fig. 1. SOC vs Time for 24-Hour Period.

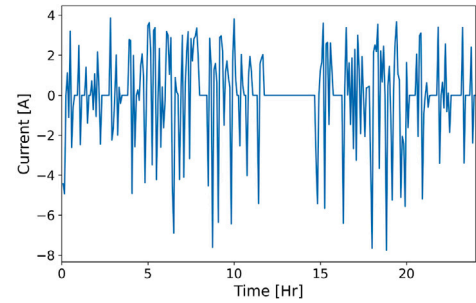


Fig. 2. Current time-series for 24-hour period.

where positive is charging and negative is discharging. The C-rate is then determined by sampling from a uniform probability distribution between 0 and either the maximum allowable charge/discharge rate ($C_{\text{max/min}}$) or the maximum C-rate that will take the battery to its boundary SOC limits ($SOC_{\text{min/max}}$) in the given step time dt . Mathematically, this is given by:

$$\text{C-rate} = \begin{cases} U(0, \min(C_{\text{max}}, \frac{SOC_{\text{max}}}{dt})), & \text{if current direction} = 1 \\ U(0, \min(C_{\text{min}}, \frac{SOC_{\text{min}}}{dt})), & \text{otherwise} \end{cases}$$

where C_{max} is 0.5, C_{min} is 1, SOC_{min} is 30%, SOC_{max} is 70%, and dt is 0.0833 h (5 min). For each time step a new current direction and C-rate is sampled. Each time step represents a 5 min constant current charge or discharge at the given C-rate. The resulting SOC and current time-series are shown for 24-hours in Figs. 1 and 2. Although the figures show only 24-hours, the actual time-series cycles uniquely between 30 and 70% SOC for 2 weeks before repeating.

3. Data-driven CNN model

The motivation for the CNN model was to map charge profiles to SOH values to see if a model could be developed that is independent of the degradation discharge/charge profile (a.k.a battery history). The CNN architecture was chosen because of its ability to map the multidimensional time-series data to a single SOH value and its proven success in similar applications in [13,14].

The data used to train the CNN was the CC and CV charge time-series from the characterization cycles, which are aligned in time such that at a SOC of zero (minimum voltage), time equals zero. The inclusion of battery temperature in the dataset was considered; however, preliminary tests with the model showed no improvement with temperature as an input. Consequently, the model operates under the reasonable assumption that the ambient temperature is controlled

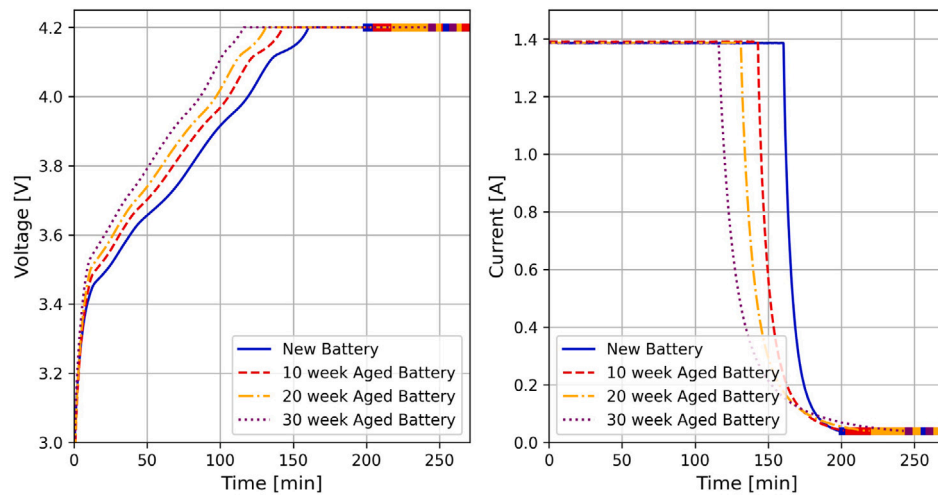


Fig. 3. Voltage and current charge profiles at new, 10 week, 20 week, and 30 week ages. Thicker line thickness indicates where data was extended.

to be 35 °C. The characterization data was selected for two reasons. First, the characterization cycle contains the most precise measurement of the SOH. Secondly, the characterization cycle is the same for all 14 batteries in the data set, hence, the hypothesis of battery usage independence can be easily tested. To increase ease of implementation and practicality of the model the data processing was kept to a minimum. First, to reduce the amount of data, the charge cycles were down sampled from a data point every 5 s to a data point every 35 s, corresponding to an increase to 0.003% SOC per step instead of 0.0004% SOC per step at the characterization cycle charge rate of 1/3C. Down-sampling in this way has no impact on the data quality because the system dynamics are slow enough that a sample every 35 s is sufficient. Next, the charge profiles were resized to fulfill the CNN constraint of consistently sized input. To ensure the consistent length of the charge cycles the non-uniform charge times were simply extended at their final values until they were the same length as the longest charge time. The data set modification only extends the constant voltage and zero current portions of the charge profiles which has a negligible effect on model accuracy because it is the least informative data. The extension of the data is illustrated in Fig. 3.

Once the charge profiles are processed they are concatenated into an input array of dimensions ($m \times n \times c$) where m is the number of characterization charge cycles, n is the length of the time series (same for every charge cycle), and c is the number of channels (which is 2 because there are voltage and current channels). Since one combined voltage and current charge profile is being mapped to one SOH value, it follows that the output is a ($m \times 1$) array of SOH values. During training and validation $m > 1$; however, when using the trained model $m = 1$ as shown in Fig. 4.

4. CNN-only model tuning

This section describes the process of tuning the CNN to obtain an optimal model. Because of the proliferation of machine learning platforms such as Tensorflow and PyTorch the implementation of common machine learning models has become relatively simple. Additionally, it is the authors' belief that a summary of the mathematics behind CNN's and a discussion of how they work is unnecessary as there are many great resources that will provide better material, including [19]. Consequently, only a brief overview of CNN concepts will be presented and the rest of this section will focus on the impact of various hyperparameters and the insight gained from analyzing the results in the specific application of lithium-ion battery SOH prediction.

In general, a CNN can be viewed as a feature extractor or detector. In the simplest sense, a CNN layer achieves feature detection by

convolving a kernel over the input, and operating on the kernel output with an activation function plus a bias. In Fig. 4 the first convolution layer has 5 randomly initialized kernels with a length of 5 that all independently sweep over the input and produce their own output, resulting in 5 filters. Following the first layer is a max pooling layer where the kernel of length 2 simply selects the maximum value within the kernel window, reducing the size of the filters by a factor of the kernel length. The rest of the model continues on similarly until it reaches the output.

Before the hyperparameters of the tuned model in Fig. 4 were determined an educated guess about the initial model architecture was required. The initial guess was to use two 1D convolution layers with max pooling between layers, a flattening layer following the CNN, and a final fully connected layer to a single SOH prediction as depicted in Fig. 4. The two CNN layer parameters were 16 and 32 filters, a sweep of 1, a kernel size of 5, and a ReLu activation function. The final fully connected layer was initialized to 64 neurons with ReLu activation functions. Next, the model was trained for 500 epochs on all of the characterization charge cycles from channels 1, 2, 4, 9, 11, 12, 13, 14, and 15 and validated on the cycles from channels 3, 10, and 16; an approximately 70/30 split. This results in 153 distinct charge profiles at varying SOH's for training. The relatively small training set provides an opportunity to test how the model handles smaller datasets characteristic of battery testing campaigns due to the long time needed for degradation. Additionally, the model is evaluated on the dynamic charge data of channels 6 and 7 to evaluate the models performance on dynamic data. The results are shown in Figs. 5 and 6 and the MAE is presented in row one of Table 3.

The prediction error is remarkably low for a first guess. Regardless, the model must still be analyzed so that areas of improvement can be identified. First, we define improvement as a reduction in prediction error or a reduction in model size while maintaining or reducing the current prediction error.

There are two likely avenues for reducing the error. First, a larger model with regularization could be built. Second, smaller models can be tested to see if the current model is over-parameterized. Due to the initial models high performance, it is unlikely that the prediction error will reduce significantly with a larger model. Additionally, it is likely that a reduction in model size (number of parameters) would be more useful in most practical situations; therefore, multiple smaller models are trained and evaluated to study the effect of hyperparameters on the model fit.

Because the initial model provided remarkable results, the general architecture is retained and only the parameters within the layers are varied. Consequently, the number of filters in the first and second

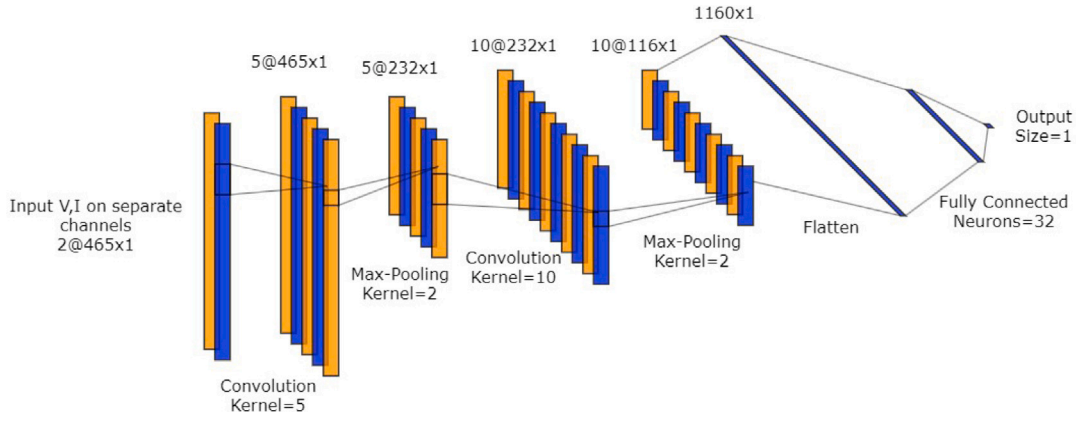


Fig. 4. Optimally tuned model dimensions shown graphically. The layer output dimensions are shown above and the layer operations are shown below.

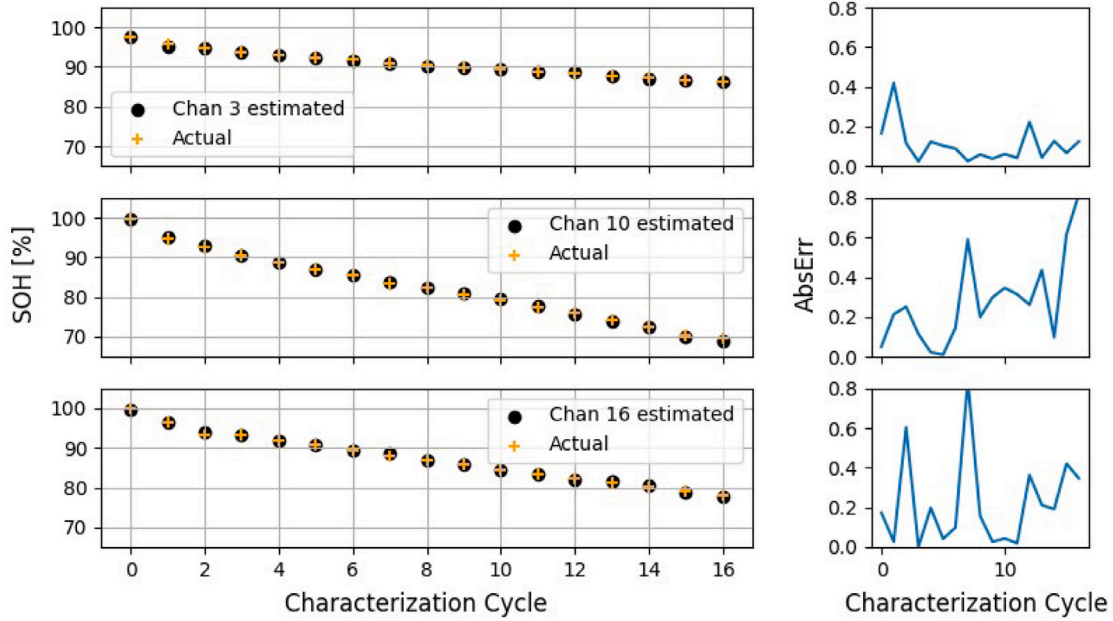


Fig. 5. Initial model SOH prediction on static validation cells. The line plot illustrates absolute error and the scatter plot illustrates predicted vs measured SOH for every characterization cycle in the validation set.

Table 3

Model prediction results for the CNN with different hyperparameters.

Parameters (filters1, filters2, Neurons, kernel)	Static MAE [SOH %]	Dynamic MAE [SOH %]	Static Max Error [SOH %]	Dynamic Max Error [SOH %]
16, 32, 64, 5	0.203	0.387	0.840	1.467
5, 10, 64, 5	0.305	0.217	2.795	0.788
2, 4, 16, 5	0.348	0.177	5.932	0.385
5, 10, 32, 5	0.217	0.278	0.880	0.847
5, 10, 32, 10	0.161	0.433	0.588	1.543
5, 10, 32, 2	0.229	0.515	1.838	1.394
5, 10, 32, 4	0.261	0.699	2.430	2.085

convolution layers, and the number of neurons are the only varied parameters. After the best combination of filters and neurons is discovered, the effect of kernel size in the first and second convolution layers is also investigated. The results in Table 3 reveal that there is not a large difference in the model performances. Regardless, it is still evident that the model with the combination shown in bold has the most consistent performance and is clearly the most generalizable to dynamic data without sacrificing static performance. The graphical results for this best combination of hyperparameters on both static and

dynamic validation data are shown in Figs. 7 and 8 and a visualization of the internal workings and layer dimensions is shown in Fig. 4.

5. Integrated gradients results analysis

Although the CNN model achieves impressive results it is impractical to assume that a full charge, from 0 to 100% SOC as defined by cut-off voltages, will be regularly encountered during a battery's life-time. In real use cases, batteries are discharged and charged arbitrarily

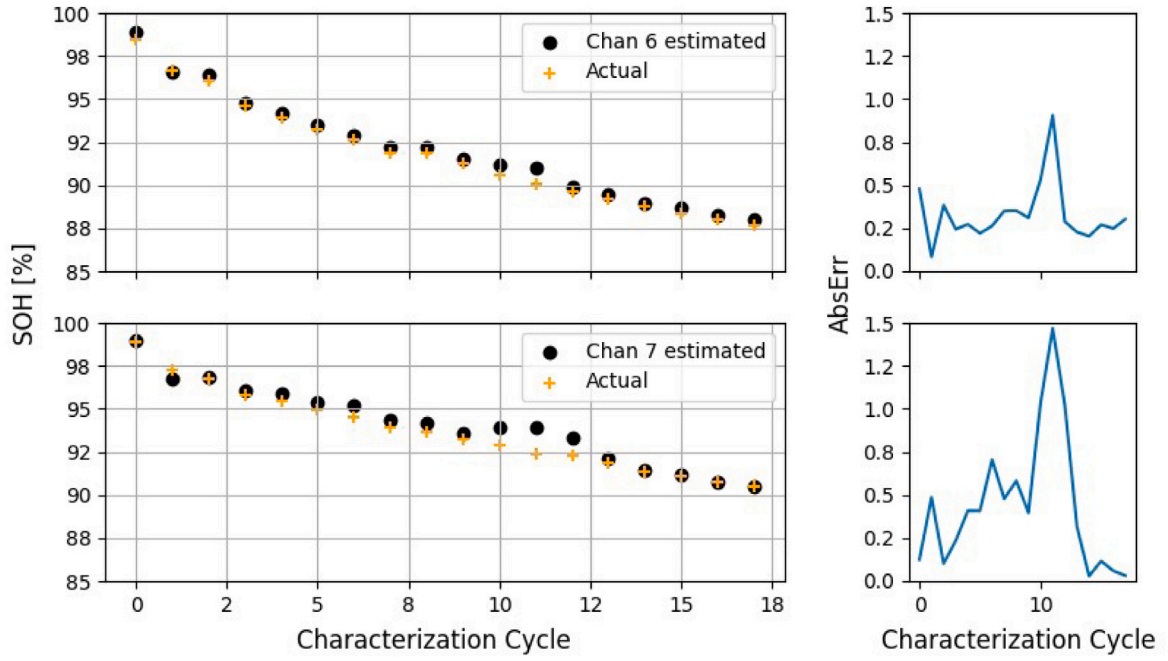


Fig. 6. Initial model SOH prediction on dynamic validation cells. The line plot illustrates absolute error and the scatter plot illustrates predicted vs measured SOH for every characterization cycle in the dynamic data set.

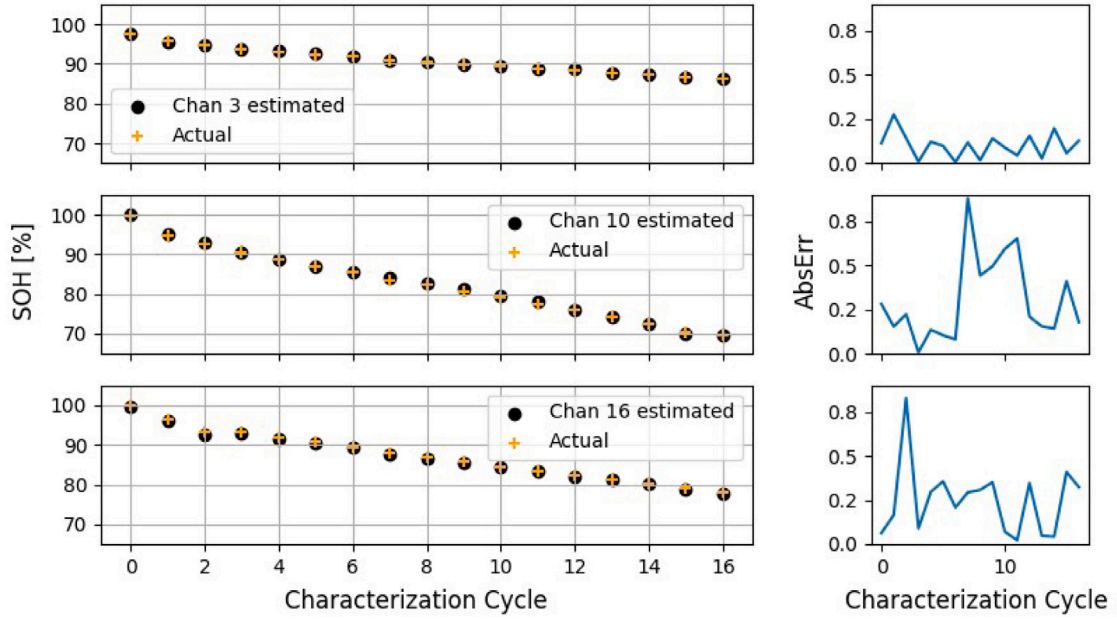


Fig. 7. Best model SOH prediction on static validation cells. The line plot illustrates absolute error and the scatter plot illustrates predicted vs measured SOH for every characterization cycle in the validation set.

over their SOC ranges. Consequently, an investigation of the models interpretation of the data was performed to understand which portions of the charge cycle are most informative. To interpret the relationship between data and model, the method of integrated gradients (IG) was used [20–22]. IG is commonly used when performing classification, especially in the case of classifying images, however, it is rarely used in regression circumstances. The method calculates the change in model prediction accuracy with respect to a data points distance from a baseline (gradient). In terms of images, this would mean the impact a pixel has on image recognition as it transitions from black (the baseline) to its final color and brightness. In terms of charge segments, the choice of baseline is much more difficult. Multiple baselines were

experimented with until the baseline of a new battery charge segment was chosen. The baseline analysis can be seen in the accompanying GitHub repository¹ where the baseline is selected to be the charge cycle of a new battery.

Once a suitable baseline is selected, the integrated gradients are calculated according to Eq. (1).

$$IG_i^{(approx)} = (x_i - x'_i) \sum_{k=1}^m \frac{\partial F(x' + \frac{k}{m}(x - x'))}{\partial x_i} \cdot \frac{1}{m} \quad (1)$$

¹ Code repository link: <https://github.com/seanbuchanan-eng/CNN-GPR>.

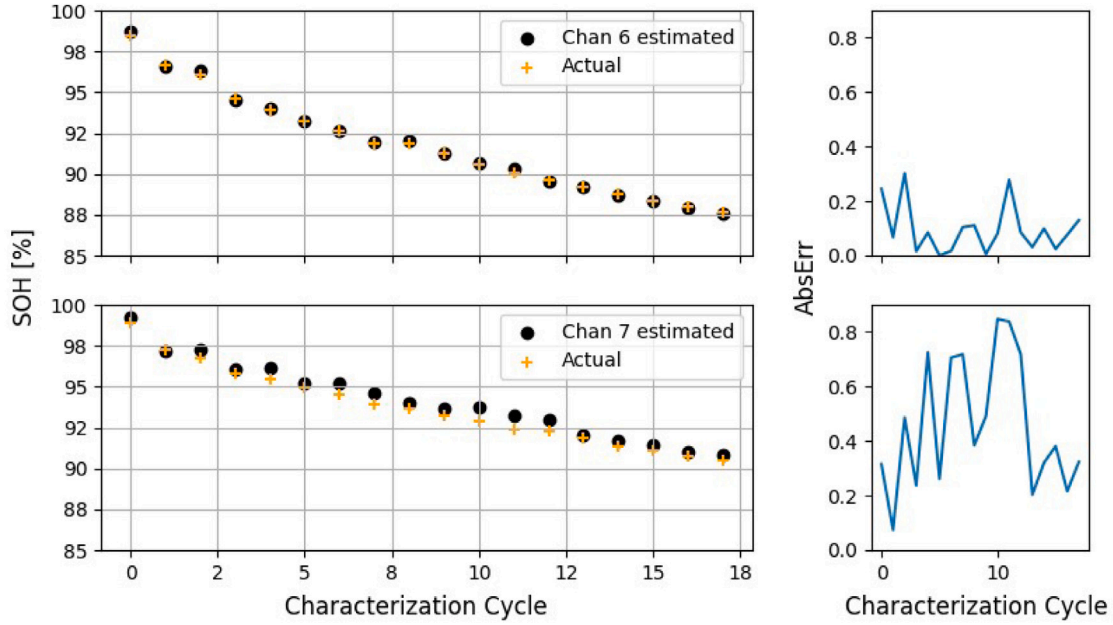


Fig. 8. Best model SOH prediction on dynamic validation cells. The line plot illustrates absolute error and the scatter plot illustrates predicted vs measured SOH for every characterization cycle in the dynamic data set.

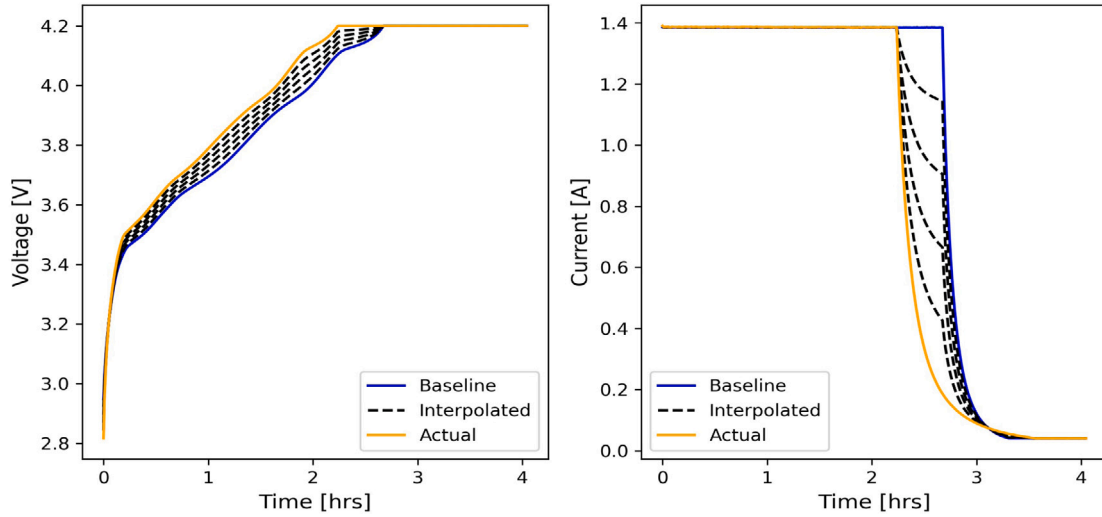


Fig. 9. Interpolated voltage and current between new battery and 18-week old battery.

where, i = feature (individual time-series data point), x = input (charge profile), x' = baseline (new battery charge cycle), k = scaled feature perturbation constant, and m = number of steps in Reimann sum approximation of the integral. Eq. (1) can be broken down into a few segments to better understand the computation. First, m interpolations are performed at k intervals between the input charge profile and the baseline. This computation corresponds to $x' + \frac{k}{m}(x - x')$, in the partial derivative in Eq. (1) and is illustrated in Fig. 9 with $m = 3$. In reality the interpolation is performed over many more steps, such as 50 in the case of this paper.

Once the interpolations are performed, the partial derivative in Eq. (1) is calculated, where $F()$ is the models prediction function and $\frac{\partial F}{\partial x_i}$ is the gradient of the models prediction function relative to each input data point. Finally, the IG attributions are calculated by integrating the gradients using the Reimann trapezoidal integral approximation. The resulting attributions represent the importance of each data point in each charge profile (voltage or current) and are plotted as a heatmap along the charge profile curves as shown in Figs. 10 and 11, where 1

in the color bar signifies most important and 0 signifies least important points.

What do the results in Figs. 10 and 11 say about the model and how it interprets the data? The most obvious observation is that the model finds the region of minimum voltage to approximately 3.7 V relatively uninformative for both voltage and current data. Next, there is a region of mild importance from 3.8 V to 4.0 V with increasing importance after 4.0 V until maximum voltage is reached. Additionally, there appears to be a vertical correlation in attribution intensity suggesting a time dependence. The time dependence indicates that the relative orientation of the charge cycles in time is important, confirming our decision at the beginning of the paper to align the charge profiles in time. Finally, there is a noticeable increase in attribution magnitude around areas of curvature suggesting that the model is gaining more insight from the nonlinear portions of the data.

It is important that the insight gained from the IG method is leveraged to improve the practicality of the model by either reducing the amount of data needed for training and predicting, improving

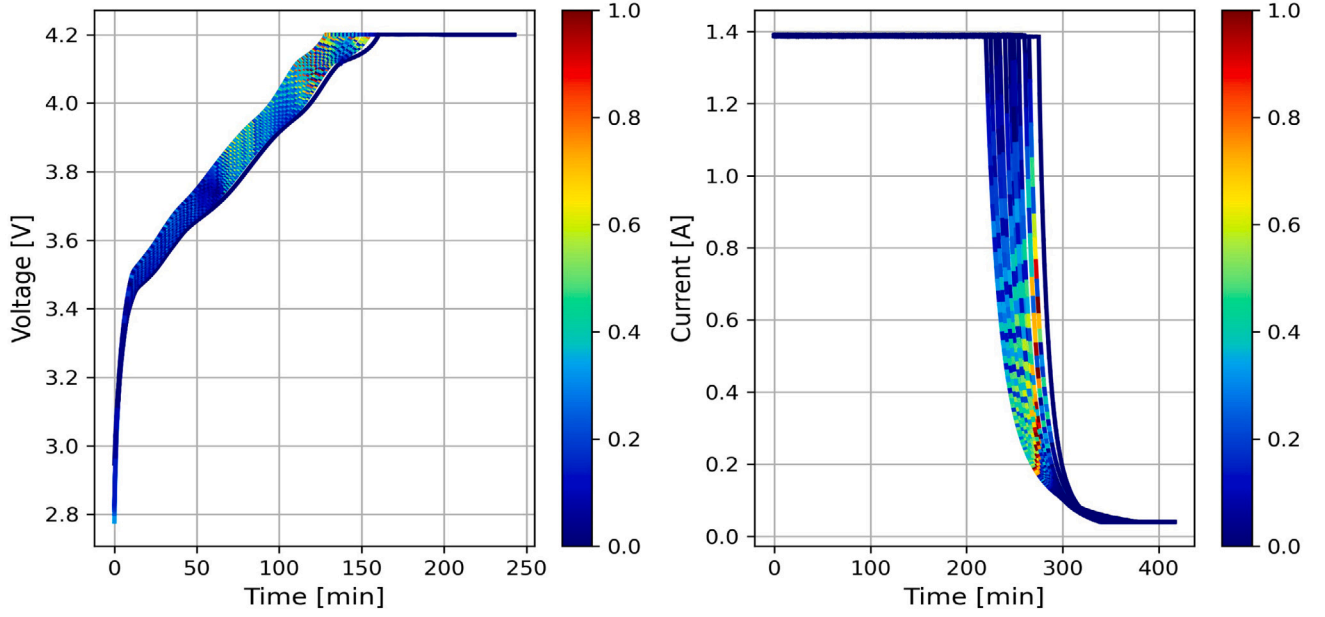


Fig. 10. Integrated Gradients attributions for cell 1 at every characterization cycle.

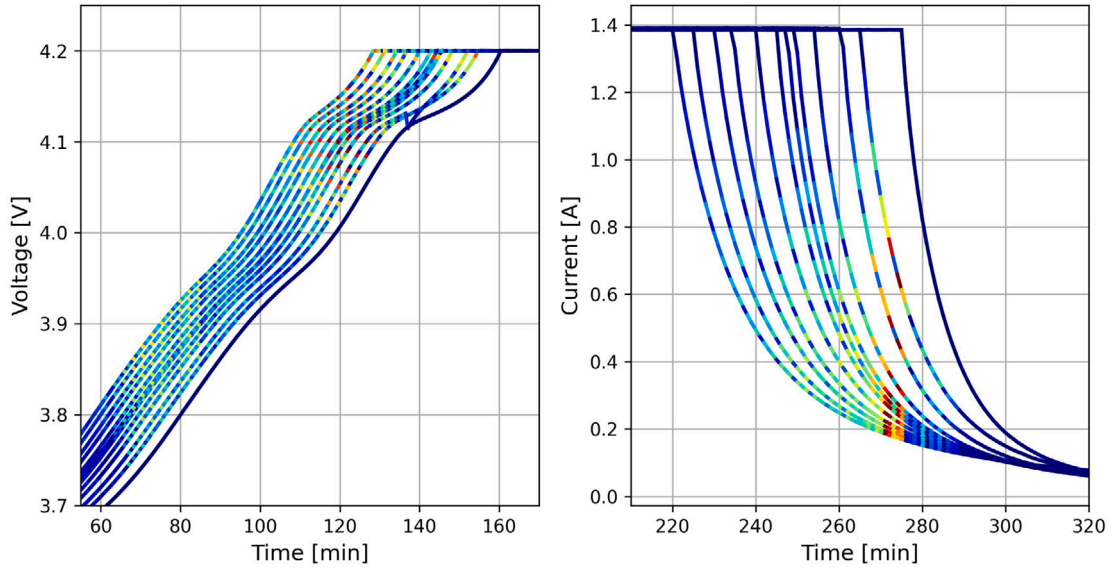


Fig. 11. Zoomed Integrated Gradients attributions for cell 1 at every characterization cycle.

the predictive performance of the model, or making the model more flexible, general, and robust. The insight gained does not support using less data because IG shows us that the most informative data is near the end of the charge cycle where the probability of it being encountered consistently is low. Improving the performance of the model is not practical either because it already has such a low prediction error. Consequently, the improvements of the model must stem from areas in flexibility, generalization, and robustness. To improve these factors, the model should ideally work on any charge segment, not just the full charge data or some specific subset of a full charge.

From the IG results it is evident that training a model on random charge segments would result in varied prediction accuracy depending on the length and location of the data in the charge cycle. The introduced randomness in the input and variance in the output suggests that a probabilistic model can be built that provides a mean and variance for the predictions based on which portion of charge is used. A natural extension of this realization is to create a combined CNN and

GPR model that provides probabilistic predictions. The probabilistic predictions will inform users of the model accuracy given what portion of charge it receives as input with the expectation that opportunistic CV portions of the charging profile will have lower uncertainty compared to CC and lower voltage segments. The creation and results of the new CNN-GPR model are described in Section 7.

6. Partial charge cycle data processing

To simulate the partial charge cycles that a battery is likely to encounter in its lifetime, additional data was created by randomly sampling from the existing characterization charge cycles. The random cycle sampling consists of three processes: removing bias, sampling from the full charge cycles, and aligning the partial charges in time. All three processes are described in the following subsections.

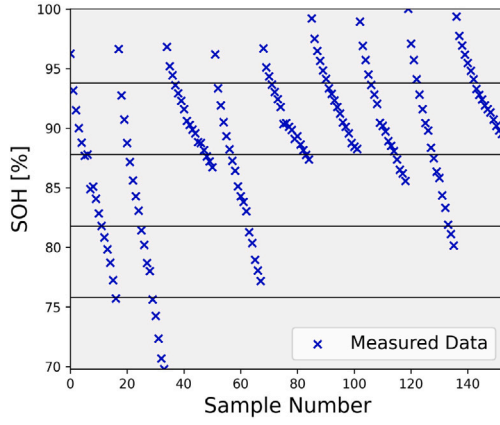


Fig. 12. Illustration of the bin boundaries represented by horizontal lines overlaid on the original unaugmented dataset.

6.1. Removing bias

The bias in the dataset stems from the fact that the batteries have different levels of final degradation after the complete testing campaign, given the range of C-rates and DOD's used across the set of batteries. Consequently, there are many more data points in the mid SOH region than the low SOH region. This bias has a negligible impact when training with full charge cycles because there are relatively few training charge profiles (approximately 150). However, when the data is augmented with duplicated charge cycles the number of charge profiles increases, hence, the bias increases. To explain the data augmenting process we define the dataset as a set of voltage and current charge profile input array's, $I_{i,j}$, and output SOH's, $SOH_{i,j}$, where i is the bin number and j is the sample number. The SOH's are then sorted into N evenly spaced bins, where $N = 5$ for this paper, and each bin has a number n_i of $SOH_{i,j}$'s in it. To remove the bias introduced by the inequality between the n_i 's, the $SOH_{i,j}$ and their corresponding input arrays $I_{i,j}$ in each under represented bin are duplicated until all n_i 's are equal. A visual representation of the bins is shown in Fig. 12.

6.2. Random sampling

The random sampling is parameterized by two variables, one global constraint, and one conditional constraint. Beginning SOC (SOC_b) and segment length in units of time (SL) are the variables of the sampling process. The global constraint constrains the segment length in time to span a change in SOC of at least 10%. The conditional constraint ensures that the beginning voltage of a charge cycle is less than 4.10 V; this constraint is conditional because it is only applied to validation data. The conditional constraint reduces the maximum prediction error by ensuring that the charge segment contains some non-constant voltage data. This approach is practically relevant as it is unlikely that the voltage will not drop below its maximum before a system is recharged.

To perform the random cycle sampling, an augmented dataset as described in Section 6.1 is produced. Then, for every charge profile $I_{i,j}$ in the dataset, a new beginning SOC (SOC_b) and length of charge (SL) is selected from discrete uniform distributions between 15 and 85 percent, and 30 and 115 min, respectively. The portion of the original full charge segment starting at SOC_b and continuing for a length of time SL is then extracted to make a new partial input charge segment $I_{pc,i,j}$. The new partial charge segment and its corresponding SOH label $SOH_{pc,i,j}$ are then added to the partial charge dataset. This process is repeated multiple times for each full charge profile, $I_{i,j}$, in the augmented dataset until a sufficient number of partial charge segments have been added to the partial charge dataset.

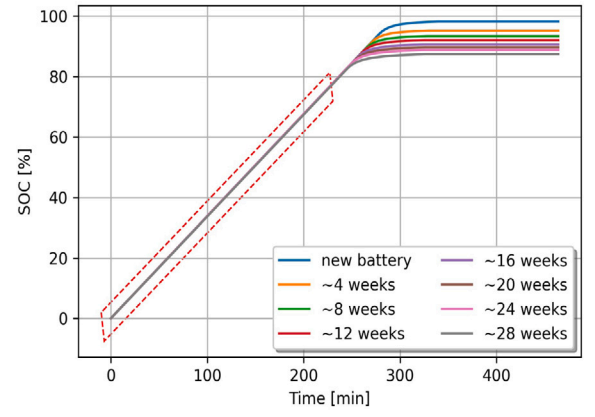


Fig. 13. SOC (defined by new battery capacity) vs time curves for channel 3 at different stages of life.

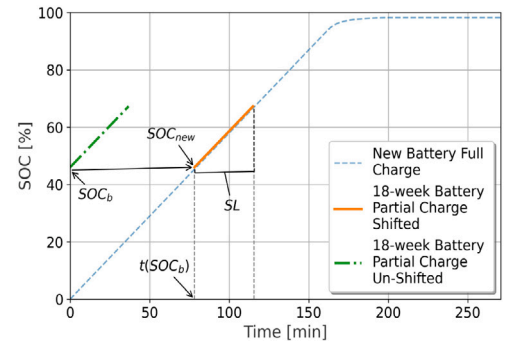


Fig. 14. Alignment of partial charge in time with new battery full charge.

6.3. Aligning partial charge segments

The IG analysis in Section 5 showed that the model relies on the charge segments being consistently aligned in time. Hence, the partial charge segments also need to be aligned in time to maintain the models prediction accuracy. In Section 3 it is stated that the full charge segments are aligned in time such that $t = 0$ at $SOC_b = 0$ for all charge cycles; however, now that $SOC_b \neq 0$, t must be shifted from zero such that the partial charge cycle maintains its position in time relative to the charge cycle of a new battery. Since SOC_b always occurs within the CC portion of the charge cycle it will always lie on the linear, highlighted, portion of Fig. 13 where all charge profiles overlap with the new battery charge cycle regardless of their SOH since the charging rate is the same in all cases.

If we define time as a function of new battery SOC ($t(SOC_{new})$) then to ensure the same position for the partial charge cycles relative to a new charge cycle, the partial charge cycles are shifted such that $SOC_{new} = SOC_b$, giving $t = t(SOC_b)$ for all partial charges. Once the partial charge is shifted, its value at the beginning and end of the segment are appended until it spans the same length of time as a full charge of a new battery. Fig. 14 illustrates the alignment in time from $t = 0$ to $t(SOC_b)$ of a partial charge cycle, and Fig. 15 demonstrates how the appended partial charge cycle remains aligned with the full charge it came from, despite being aligned with SOC_{new} from the full charge cycle.

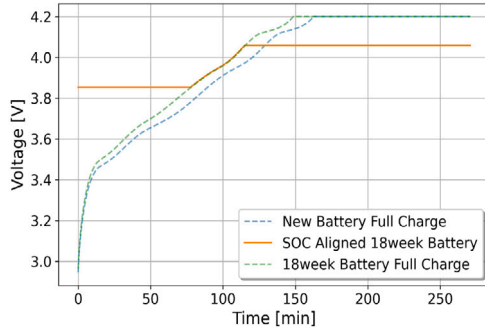
This method of alignment relies on two assumptions; first, that there is a reliable model estimating SOC and second, that there is a full charge cycle, from 0 to 100% SOC, when the battery is new. Both assumptions are reasonable given the advanced battery management systems in use today and the fact that new batteries undergo a full charge at beginning of life.

Table 4Model prediction results for the CNN-GPR from 100 evaluations with an operating range of 15%–85% SOH_b .

	MAE_{100} [SOH%]	$MaxE_{100}$ [SOH%]	Max Error [SOH%]
Static Validation Results	0.7	3.6	10.9
Dynamic Validation Results	0.7	4.1	15.3

Table 5Model prediction results for the CNN-GPR from 100 evaluations with an operating range of 30%–80% SOH_b .

	MAE_{100} [SOH%]	$MaxE_{100}$ [SOH%]	Max Error [SOH%]
Static Validation Results	0.6	2.3	4.4
Dynamic Validation Results	0.6	2.1	3.3

**Fig. 15.** Partial charge overlay with original full charge cycle and new battery full charge cycle.

7. CNN-GPR model

As with the standalone CNN model, explaining the mathematical nuances of the CNN and GPR model formulations are left to machine-learning Refs. [19,23]. The GPR will only be summarized and then the model construction, application, and results will be presented in this section.

7.1. GPR overview

In general, when trying to map an input–output relationship using data there are infinite possible functions that can be selected. Gaussian process regression uses Bayesian inference with a kernel function as a prior to find a distribution of functions that fit the data with a mean function that has the highest probability of describing the data.

Mathematically, if the CNN output is the estimated SOH, $\widehat{SOH}_{CNN}$, then

$$\widehat{SOH}_{CNN-GPR} = f(\widehat{SOH}_{CNN}) \quad (2)$$

where $f()$ is the GPR function and $\widehat{SOH}_{CNN-GPR}$ is a Gaussian distribution calculated using $\widehat{SOH}_{CNN}$ and a kernel [24].

Because the Gaussian process is probabilistic it also provides an estimate of the models confidence in its prediction. In the CNN-GPR, the Gaussian process regression relies on Sparse Variational Gaussian Process' (SVGP) as defined in the gpflux package [24]. The use of the SVGP algorithm reduces the computational burden of traditional Gaussian process regression by using inducing points. Additionally, because the GPR in the CNN-GPR is being trained on the single output of the CNN instead of the original time-series charge cycles there is a reduction in training points, further reducing training times when compared to a stand alone GPR model. The final CNN-GPR model consists of the CNN model architecture and best parameters presented in Section 4, combined with a GPR layer as shown in Fig. 16.

7.2. CNN GPR interaction

Intuitively, one might expect the CNN and GPR to act sequentially such that the CNN predicts SOH as accurately as possible and then the GPR fits the CNN estimation to correct for the varying CNN SOH estimations. Fundamentally, this understanding is the same as if the CNN and GPR are trained separately. Fig. 17 shows the resulting fit of a linear kernel GPR trained with the stand alone CNN's estimated SOH as input and the measured SOH as output.

Alternatively, if the GPR is added as a final layer to the CNN such that the CNN and GPR become a single model that is trained as one, the CNN then becomes a transformer for the GPR. The combined configuration now causes the CNN to transform its output relative to the stand alone CNN such that the GPR has the best form of inputs for its kernel function. Consequently, if there is an expected form for the CNN output, such as the linear relationship between estimated and measured SOH's, then the GPR kernel choice can be used to inform the CNN of the best way to transform its output to match the expected output. Effectively, the GPR kernel is no longer just a tool to fit a given input, but a tool to inform a transformer of the most effective basis coordinates for accurate GPR estimation. Figs. 17–20 compare the CNN GPR interaction of a combined CNN-GPR with its independently trained counterpart for linear and squared exponential kernel selections. The figures show the models fit to the same training data used throughout the paper.

Figs. 19 and 20 provide a noticeable shift in model behavior between independent and combined CNN and GPR, however, it is the differences between the $\widehat{SOH}_{CNN}$ axes in Figs. 17 and 18 that is most notable. The $\widehat{SOH}_{CNN}$ axis in Fig. 18 shows how the CNN has compressed the output space to increase linearity and reduce the variance in its output. As expected, the model that provides the lowest mean average error, mean squared error, and max error is the linear kernel with the fully connected CNN-GPR architecture. Consequently, the fully connected CNN-GPR architecture with a linear kernel is the chosen model and its performance and results will be discussed in the following section.

7.3. CNN-GPR results and discussion

The CNN-GPR training dataset contains the same 153 full characterization charge segments from channels 1, 2, 4, 9, 11, 12, 13, 14, and 15 as the original CNN, plus an additional 1,957 partial charge segments sampled from the full characterization charge segments of the cells just mentioned, equaling a total of 2,110 training charge cycles. The validation dataset is split into two subsets; the first contains data from the CC discharge (static) cycle tests from channels 3, 10 and 16, and the second contains data from the variable C-rate (dynamic) cycle tests from channels 6 and 7. The same process was followed as described in Section 6 to obtain partial charge segments with one difference compared to the training set. The validation dataset only contains one partial charge sample ($Ipc_{i,j}$) for every SOH label while the training dataset contains multiple. The model was trained on an AMD Ryzen 5 5600H 3.3 GHz CPU with 16 GB of RAM for 200 epochs with a duration

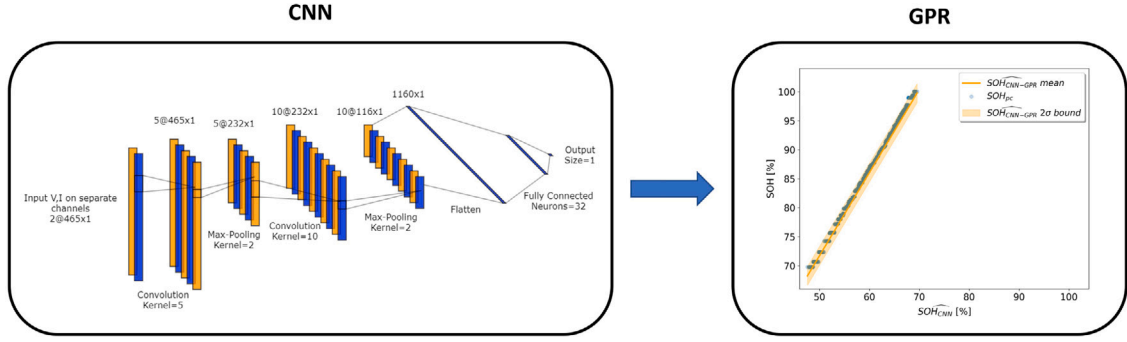


Fig. 16. CNN-GPR architecture. GPR operates on the CNN estimate and both submodels are embedded in the same model such that the GPR and CNN are trained together.

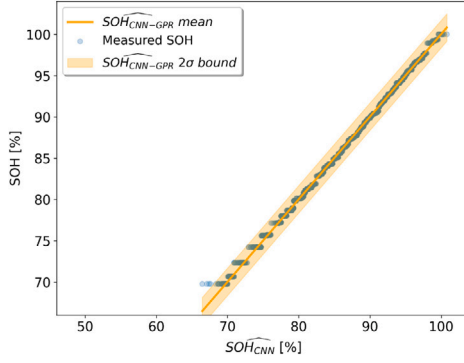


Fig. 17. Independently trained CNN and GPR with linear kernel.

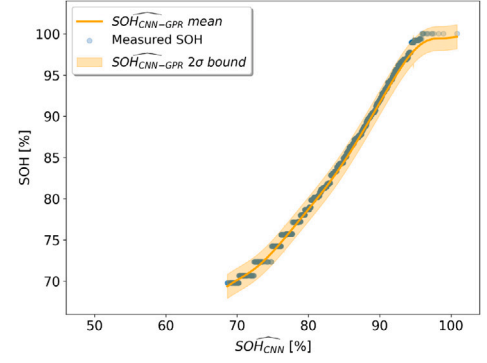


Fig. 20. Combined CNN-GPR with squared exponential kernel.

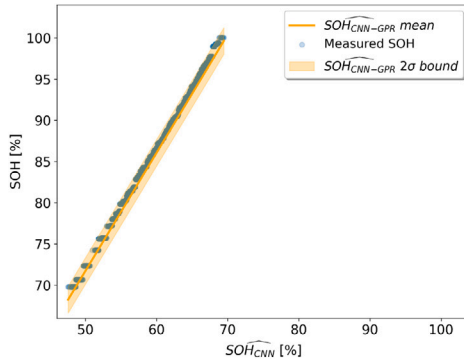


Fig. 18. Combined CNN-GPR with linear kernel.

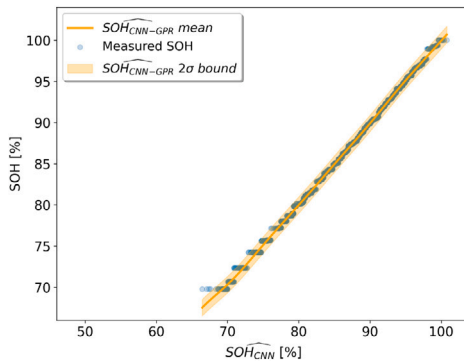


Fig. 19. Independently trained CNN and GPR with squared exponential kernel.

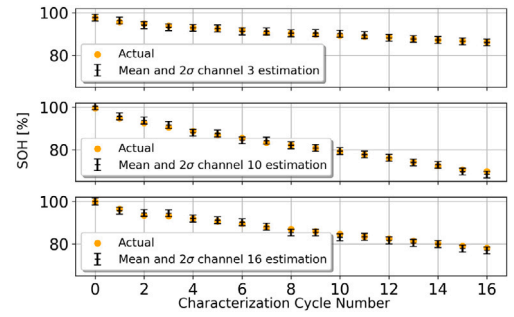


Fig. 21. Resulting fit for one set of static validation $Ipc_{i,j}$ inputs. MAE: 0.6%, MaxE: 2.3%.

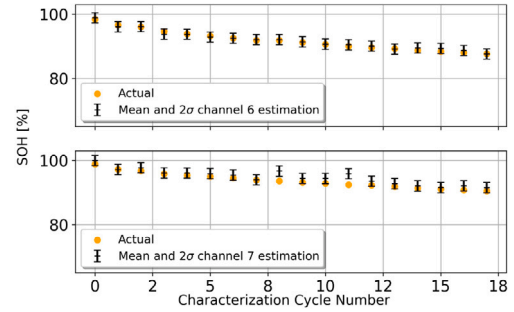


Fig. 22. Resulting fit for one set of dynamic validation $Ipc_{i,j}$ inputs. MAE: 0.6%, MaxE: 2.1%.

of 1 min and 22 s. An example result for every SOH label in the static validation set is shown in Fig. 21 and for the dynamic validation set in Fig. 22.

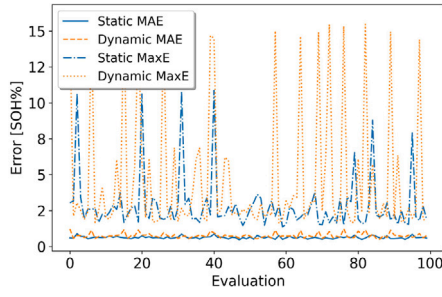


Fig. 23. Error metrics for each of the 100 model evaluations using newly sampled $I_{pc_{i,j}}$'s from the static validation dataset.

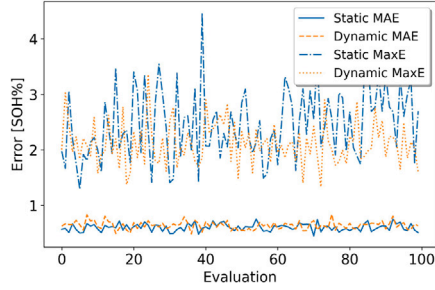


Fig. 24. Error metrics for each of the 100 model evaluations using newly sampled $I_{pc_{i,j}}$'s from the dynamic validation dataset.

Because the validation dataset contains only one partial charge segment for every SOH label, it is not sufficient to characterize the models performance. Consequently, the model's performance was evaluated by re-sampling a new $I_{pc_{i,j}}$ for every SOH label in the validation dataset and then running the model with the new $I_{pc_{i,j}}$'s. The predicted SOH's are then evaluated by calculating the MAE and maximum error (MaxE) between the predicted SOH's and the measured SOH's. This process is then repeated 100 times to produce two aggregate error metrics, MAE_{100} and $MaxE_{100}$. MAE_{100} is the average of the MAE's over all 100 model evaluations and $MaxE_{100}$ is the average MaxE's over all 100 model evaluations. The resulting aggregate error metrics are shown in Table 4.

Although the static and dynamic MAE_{100} and $MaxE_{100}$ results are more than sufficient for the model to be applicable, the maximum errors seen in Fig. 23 of approximately 15% dramatically reduce the models usability. The large errors are likely caused by edge cases such as absent variable voltage data or short charges around low SOC like charging from 15 to 25% SOC. To reduce the maximum error, the model is given a reduced operating range. If the operating range limits the beginning SOC (SOC_b) of a partial charge to a range of 30%–80% then the maximum error dramatically reduces, illustrated in Fig. 24 and Table 5. The prediction results for one evaluation using the new operating range are displayed in Figs. 25 and 26.

More important than the overall error is the similar error resulting from both the static and dynamic validation datasets, despite the model only being trained on the static data. This result suggests that the CNN-GPR accurately predicts battery SOH regardless of how the battery is cycled for degradation, and is only reliant on periodic and opportunistic charging cycle segments for effective SOH estimation.

8. Conclusions

This paper investigated the input data/prediction relationship of a CNN predicting state of health for lithium-ion batteries using the Integrated Gradients method. The Integrated Gradients method revealed that mid-voltage, high-voltage, and constant voltage charge segments

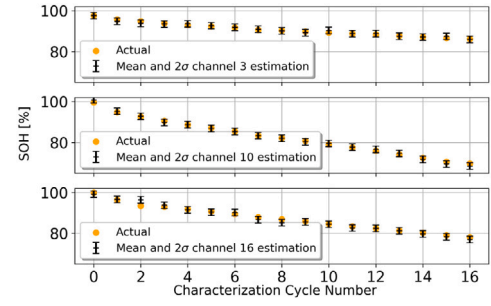


Fig. 25. Resulting fit for one set of static validation $I_{pc_{i,j}}$ inputs using the SOC_b operating range of 30%–80%. MAE: 0.6%, MaxE: 2.7%.

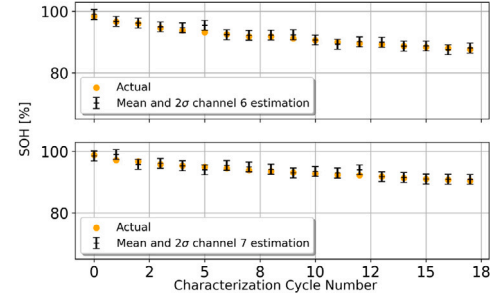


Fig. 26. Resulting fit for one set of dynamic validation $I_{pc_{i,j}}$ inputs using the SOC_b operating range of 30%–80%. MAE: 0.5%, MaxE: 2.1%.

are all strong predictors of battery state of health. This information implies that input of randomized charge profile lengths and beginning state of charge values possess a variance based on which portion of the charge profile is used. Consequently, the CNN-GPR model was constructed to capture this variance and provide a probabilistic prediction of battery state of health. The probabilistic prediction is especially useful where partial charges are common and accurate battery lifetime prognostics are required. Finally, the CNN-GPR performance is tested on both static and dynamic battery usage data. The CNN-GPR predicts the battery state of health with less than 0.7% average MAE and 4.5% maximum absolute error over 100 randomly sampled partial input charges on dynamic data. Remarkably, the CNN-GPR achieves these results when it has only been trained on static test data.

To extend the predictive success of the CNN-GPR there are two main elements that can be improved:

- Validate the model on a separate data set to assess its applicability to different battery chemistry's and sizes, potentially utilizing transfer learning.
- This first version of the CNN-GPR was trained and validated with uniform C-rate charge profiles and constant ambient temperature. To improve the practicality of the model, multiple C-rates and ambient temperatures should be used for training and validation.

With these small changes, this model approach will provide an opportunity to gather fleet-level insights into SOH based on variable duty and charge cycles across the fleet and for microgrids with variable load, genset, and renewable generation profiles.

CRedit authorship contribution statement

Sean Buchanan: Conceptualization, Methodology, Software, Validation, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Curran Crawford:** Conceptualization, Methodology, Resources, Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- [1] H. Tian, P. Qin, K. Li, Z. Zhao, A review of the state of health for lithium-ion batteries: Research status and suggestions, *J. Clean. Prod.* 261 (2020) 120813, <http://dx.doi.org/10.1016/j.jclepro.2020.120813>, URL <https://www.sciencedirect.com/science/article/pii/S095965262030860X>.
- [2] P. Singh, C. Chen, C.M. Tan, S.-C. Huang, Semi-empirical capacity fading model for SoH estimation of Li-ion batteries, *Appl. Sci.* 9 (15) (2019) 3012, <http://dx.doi.org/10.3390/app9153012>, Number: 15 Publisher: Multidisciplinary Digital Publishing Institute, URL <https://www.mdpi.com/2076-3417/9/15/3012>.
- [3] N. Lotfi, J. Li, R.G. Landers, J. Park, Li-ion battery state of health estimation based on an improved single particle model, in: 2017 American Control Conference, ACC, (ISSN: 2378-5861) 2017, pp. 86–91, <http://dx.doi.org/10.23919/ACC.2017.7962935>.
- [4] M. Montaru, S. Fiette, J.-L. Koné, Y. Bultel, Calendar ageing model of Li-ion battery combining physics-based and empirical approaches, *J. Energy Storage* 51 (2022) 104544, <http://dx.doi.org/10.1016/j.est.2022.104544>, URL <https://www.sciencedirect.com/science/article/pii/S2352152X22005618>.
- [5] Y. Jiang, J. Zhang, L. Xia, Y. Liu, State of health estimation for Lithium-ion battery using empirical degradation and error compensation models, *IEEE Access* 8 (2020) 123858–123868, <http://dx.doi.org/10.1109/ACCESS.2020.3005229>, Conference Name: IEEE Access.
- [6] C. Weng, Y. Cui, J. Sun, H. Peng, On-board state of health monitoring of lithium-ion batteries using incremental capacity analysis with support vector regression, *J. Power Sources* 235 (2013) 36–44, <http://dx.doi.org/10.1016/j.jpowsour.2013.02.012>, URL <https://www.sciencedirect.com/science/article/pii/S0378775313002668>.
- [7] X. Han, L. Lu, Y. Zheng, X. Feng, Z. Li, J. Li, M. Ouyang, A review on the key issues of the lithium ion battery degradation among the whole life cycle, *eTransportation* 1 (2019) 100005, <http://dx.doi.org/10.1016/j.etrans.2019.100005>, URL <https://www.sciencedirect.com/science/article/pii/S2590116819300050>.
- [8] V. Klass, M. Behm, G. Lindbergh, A support vector machine-based state-of-health estimation method for lithium-ion batteries under electric vehicle operation, *J. Power Sources* 270 (2014) 262–272, <http://dx.doi.org/10.1016/j.jpowsour.2014.07.116>, URL <https://www.sciencedirect.com/science/article/pii/S0378775314011707>.
- [9] D. Zhou, B. Wang, Battery health prognosis using improved temporal convolutional network modeling, *J. Energy Storage* 51 (2022) 104480, <http://dx.doi.org/10.1016/j.est.2022.104480>, URL <https://www.sciencedirect.com/science/article/pii/S2352152X22005023>.
- [10] S. Jo, S. Jung, T. Roh, Battery state-of-health estimation using machine learning and preprocessing with relative state-of-charge, *Energies* 14 (21) (2021) 7206, <http://dx.doi.org/10.3390/en14217206>, Number: 21 Publisher: Multidisciplinary Digital Publishing Institute, URL <https://www.mdpi.com/1996-1073/14/21/7206>.
- [11] Z. Deng, X. Hu, P. Li, X. Lin, X. Bian, Data-driven battery state of health estimation based on random partial charging data, *IEEE Trans. Power Electron.* 37 (5) (2022) 5021–5031, <http://dx.doi.org/10.1109/TPEL.2021.3134701>, Conference Name: IEEE Transactions on Power Electronics.
- [12] R.R. Richardson, C.R. Birkel, M.A. Osborne, D.A. Howey, Gaussian process regression for in situ capacity estimation of lithium-ion batteries, *IEEE Trans. Ind. Inform.* 15 (1) (2019) 127–138, <http://dx.doi.org/10.1109/TII.2018.2794997>, Conference Name: IEEE Transactions on Industrial Informatics.
- [13] E. Chemali, P.J. Kollmeyer, M. Preindl, Y. Fahmy, A. Emadi, A convolutional neural network approach for estimation of Li-ion battery state of health from charge profiles, *Energies* 15 (3) (2022) 1185, <http://dx.doi.org/10.3390/en15031185>, Number: 3 Publisher: Multidisciplinary Digital Publishing Institute, URL <https://www.mdpi.com/1996-1073/15/3/1185>.
- [14] S. Shen, M. Sadoughi, X. Chen, M. Hong, C. Hu, A deep learning method for online capacity estimation of lithium-ion batteries, *J. Energy Storage* 25 (2019) 100817, <http://dx.doi.org/10.1016/j.est.2019.100817>, URL <https://www.sciencedirect.com/science/article/pii/S2352152X19302233>.
- [15] H. Xu, L. Wu, S. Xiong, W. Li, A. Garg, L. Gao, An improved CNN-LSTM model-based state-of-health estimation approach for lithium-ion batteries, *Energy* 276 (2023) 127585, <http://dx.doi.org/10.1016/j.energy.2023.127585>, URL <https://www.sciencedirect.com/science/article/pii/S0360544223009799>.
- [16] D. Li, L. Yang, Remaining useful life prediction of lithium battery based on sequential CNN-LSTM method, *J. Electrochem. Energy Convers. Storage* 18 (041005) (2021) <http://dx.doi.org/10.1115/1.4050886>.
- [17] Y. Zheng, J. Hu, J. Chen, H. Deng, W. Hu, State of health estimation for lithium battery random charging process based on CNN-GRU method, *Energy Rep.* 9 (2023) 1–10, <http://dx.doi.org/10.1016/j.egy.2022.12.093>, URL <https://www.sciencedirect.com/science/article/pii/S2352484722026944>.
- [18] Product Data Sheet, No. INR-21700-P42A, Molicel, 2022.
- [19] U. Michelucci, *Advanced Applied Deep Learning*, A Press, Berkeley, CA, 2019.
- [20] M. Sundararajan, A. Taly, Q. Yan, Axiomatic attribution for deep networks, 2017, <http://dx.doi.org/10.48550/arXiv.1703.01365>, arXiv:1703.01365 [cs].
- [21] D. Kelly, *Integrated gradients notebook*, 2020, URL https://github.com/GoogleCloudPlatform/training-data-analyst/tree/master/blogs/integrated_gradients.
- [22] TensorFlow, *Integrated gradients*, 2022, URL https://www.tensorflow.org/tutorials/interpretability/integrated_gradients.
- [23] C.E. Rasmussen, C.K.I. Williams, *Gaussian Processes for Machine Learning*, The MIT Press, 2005, <http://dx.doi.org/10.7551/mitpress/3206.001.0001>, URL <https://direct.mit.edu/books/book/3230/Gaussian-Processes-for-Machine-Learning>.
- [24] V. Dutordoir, H. Salimbeni, E. Hambro, J. McLeod, F. Leibfried, A. Artemev, M. van der Wilk, J. Hensman, M.P. Deisenroth, S.T. John, GPflux: A library for deep Gaussian processes, 2021, <http://dx.doi.org/10.48550/arXiv.2104.05674>, arXiv:2104.05674 [cs, stat].